SME307 Analog CMOS Integrated Circuit Design

Design Project: **Op-amp Design**

Due Date: **Monday, Week 17th**

The objective of this project is to design a differential input single-ended output CMOS op-amp for a specific application where the op-amp will be used as a buffer as shown in Fig. 1. For this design only NMOS, PMOS transistors, capacitors, and resistors can be used. For this design, you are asked to use the 0.18 μm CMOS process (gpdk in the library of virtuoso):

In your op-amp design, ideal sources **cannot** be used **except** for V_{DD} and V_{SS} (i.e., ideal sources cannot be used to generate bias currents or voltages, however, it is suggested that in the first pass of your design, you use ideal bias current sources and then once you met the specifications replace the bias sources with transistor-level designs). The unity gain buffer of Fig. 1 is intended for either input step signals or sinusoidal signals with frequencies less than 60 MHz (Note that you are asked to over-design for the unity gain frequency of the op-amp). The design specifications are summarized in Table I.

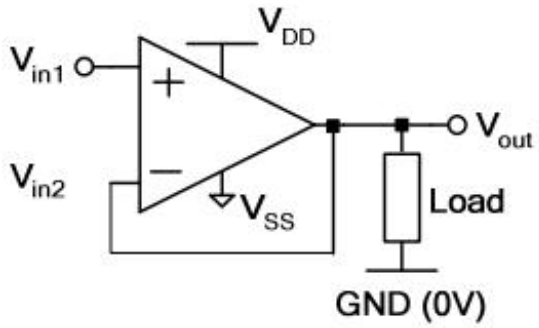


Fig. 1. Op-amp in unity-gain feedback (i.e., unity buffer)

TABLE I. DESIGN SPECIFICATIONS

Process	0.18 μm
V_{DD}	1.8 V
V_{SS}	0 V
Load Cap	1 pF
A_{DM0} (low-frequency open-loop small-signal gain)	≥ 1000 V/V
A_{CM0} (low-frequency open-loop small-signal common-mode gain)	≤ 0.1 V/V
Phase margin	≥ 60°
Unity gain frequency	≥ 100 MHz
Slew rate	≥ 10 V/μs
Output voltage swing (differential peak to peak)	≥ 800 mV
Power consumption	Minimum

Bonus:
Design your op-amp such that the specifications are met under a ±10% variation of the supply voltage. To submit your project report and the schematic of your design, please follow these steps:

Report Respect:
1. You need to hand in a Word or PDF document that includes your name, student number, a table summarizing the performance results of your op-amp, the summary of your hand calculations, and a brief description of your design approach, required plots including the bode plot of open-loop transfer function on which achieved phase margin and low-frequency gain are annotated, bode plot of closed-loop system on which the 3-dB frequency of the system is indicated, and any comments and conclusions. Please also include the schematic of your design with transistor sizes and bias currents indicated on the schematic beside each transistor (component values should be indicated beside each component).

2. Any other supporting document(s) or graph(s) that you would like to hand in should be provided. Give citations or quotations of any literature or websites that have helped you in the "Reference" part. For example:

REFERENCE

[1] Author A, Author B, Author C, and Author D, "Title of the paper," *Journal Name*, vol. XX, no. X, pp. XX-XX, Month Year.

3. It is expected to use the format requirements of IEEE journal papers in your reports. we recommend the template of Journal of Solid State (JSSC) or Transactions on Circuit and Systems I: Regular Papers (TCAS-I). If you want to find the template of journals, letters, magazines, and conferences in IEEE, you can search this website: https://template-selector.ieee.org/secure/templateSelector/publicationType

Good Luck!